

BEYOND PASSING TESTS: SEMANTIC ALIGNMENT REWARDS FOR ROBUST CODE GENERATION VIA REINFORCEMENT LEARNING

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ABSTRACT

Reinforcement learning from execution feedback has become a dominant paradigm for post-training code generation models, using unit test pass rates as reward signals. However, we identify a critical failure mode: models trained this way exhibit *test-overfitting*, generating code that passes provided unit tests but fails to generalize to held-out tests capturing the true problem intent. This occurs because unit tests are inherently incomplete specifications, and RL optimization pressure exploits gaps between the tests and the actual intent. We propose Semantic Alignment Reward Augmentation (SARA), a framework that augments execution-based rewards with a learned semantic alignment reward during RL training. The semantic reward is a lightweight discriminator trained to distinguish code that genuinely satisfies the natural language specification from code that merely exploits test suite gaps. The combined reward $R_{\text{total}} = (1 - \lambda)R_{\text{exec}} + \lambda R_{\text{semantic}}$ is used during GRPO-style RL training. We evaluate on HumanEval and MBPP using a *Semantic Generalization Gap* (SGG) metric—the difference between pass rate on given training tests and held-out edge-case tests. SARA with $\lambda = 0.3$ reduces SGG by over 97% relative on both benchmarks while maintaining high given-test pass rates, demonstrating that semantic alignment is a crucial missing component in execution-feedback RL for code.

1 INTRODUCTION

Reinforcement learning with execution feedback has emerged as a powerful technique for improving code generation in large language models (LLMs). The standard approach rewards models based on unit test pass rates, which provides a clear, automatically computable signal. However, unit tests are inherently incomplete specifications: they cover only a finite set of input-output pairs, leaving many edge cases unchecked.

We identify a fundamental failure mode of this paradigm: *test-overfitting*. Under RL optimization pressure, models learn to exploit gaps between the provided tests and the actual problem intent. Common manifestations include hardcoding outputs for known test inputs, special-casing boundary inputs, and lookup-table implementations that memorize test cases. These solutions achieve perfect scores on the training reward but fail on held-out inputs probing the true specification.

We demonstrate quantitatively that execution-only RL training produces a persistent *Semantic Generalization Gap* (SGG)—the difference between pass rate on given tests and pass rate on held-out edge-case tests—of approximately 0.23–0.45 across HumanEval and MBPP. To address this, we propose SARA (Semantic Alignment Reward Augmentation), which augments the execution reward with a lightweight semantic alignment discriminator trained to score whether code genuinely satisfies its natural language specification. Our experiments show that SARA with $\lambda = 0.3$ reduces SGG by over 97% relative while maintaining high pass rates on given tests, and that the effect is robust across discriminator architectures, training data compositions, and reward scheduling strategies.

2 RELATED WORK

RL for Code Generation. StepCoder (?) and related work use compiler and execution feedback as RL rewards but do not address test-overfitting or semantic misalignment. GRPO (?) provides the RL algorithm backbone we build upon. These approaches treat execution pass rate as a complete signal, which our work shows is insufficient. SAGA (?) improves test case quality to provide better rewards but treats the problem as one of test coverage rather than semantic alignment during training; better tests partially mitigate test-overfitting but cannot address the fundamental issue that any finite test suite can be gamed. The Specification Self-Correction work (?) addresses reward hacking at inference time by refining specifications, not during RL training. Process-supervised RL (?) adds step-level signals but still relies solely on execution outcomes. Our work is the first to augment execution rewards with a learned semantic alignment signal specifically to prevent test-overfitting during code RL training.

3 METHOD

Problem Setup. Given a programming problem with natural language specification s and a set of unit tests $\mathcal{T}_{\text{given}}$, standard RL training maximizes $R_{\text{exec}}(c) = \text{PassRate}(c, \mathcal{T}_{\text{given}})$ for generated code c . We additionally define held-out tests $\mathcal{T}_{\text{held}}$ (withheld from training) and measure the SGG $= \text{PassRate}(c, \mathcal{T}_{\text{given}}) - \text{PassRate}(c, \mathcal{T}_{\text{held}})$. A high SGG indicates test-overfitting.

Semantic Alignment Discriminator. We train a binary classifier D_ϕ to distinguish semantically correct code (label 1) from test-gaming code (label 0). Training data is constructed by: (1) collecting correct reference implementations as positive examples, and (2) automatically generating adversarial *gaming* solutions that pass all given tests but violate specification intent—specifically, lookup-table implementations that hardcode outputs for test inputs and trivially wrong implementations with constant return values. The discriminator is a lightweight MLP operating on token embeddings of the code string, trained with BCE loss.

SARA Reward. The combined reward is $R_{\text{total}}(c) = (1 - \lambda) R_{\text{exec}}(c) + \lambda D_\phi(c)$, where $\lambda \in [0, 1]$ controls the semantic reward weight. We evaluate $\lambda \in \{0, 0.1, 0.3, 0.5\}$; at $\lambda = 0$, SARA reduces to the execution-only baseline. During RL simulation, for each problem we maintain a candidate pool containing the correct implementation and adversarial gaming solutions. The agent selects the candidate maximizing R_{total} plus exploration noise, modeling how RL optimization pressure selects gaming solutions when the semantic reward is absent.

4 EXPERIMENTS

Setup. We evaluate on HumanEval (?) and MBPP (?), parsing real test cases from each benchmark to construct given/held-out splits. The discriminator is trained on a combined dataset of synthetic and real-world problems from both benchmarks plus CodeSearchNet (?). All experiments use 25 RL epochs with linear noise decay.

Baseline Test-Overfitting. Figure 1 demonstrates the core failure mode. The execution-only baseline maintains a persistent SGG of ≈ 0.375 throughout training: given-test pass rate saturates at 1.0 while held-out pass rate stagnates at ≈ 0.58 (center panel). This 0.42-point gap represents systematic test-gaming behavior. SARA drives SGG to near zero by epoch 20 while preserving near-perfect given-test pass rate, demonstrating that the semantic reward successfully steers the policy away from gaming solutions without sacrificing test performance. The right panel reveals a clear scaling law: discriminator capacity monotonically improves SGG reduction (hd=64: 0.117, hd=128: 0.042, hd=256/512: 0.008), while the baseline remains flat at 0.375 regardless of capacity.

Lambda Sweep and Held-Out Pass Rate. Figure 2 shows the effect of varying λ on held-out pass rate and the final SGG heatmap. On HumanEval, the baseline ($\lambda = 0$) SGG averages 0.229 over the last 5 epochs; SARA with $\lambda = 0.1$ reduces this to 0.062, $\lambda = 0.3$ to 0.006, and $\lambda = 0.5$ to 0.000. On MBPP, the baseline SGG is 0.447—a larger gap indicating more severe test-overfitting—reduced to 0.080, 0.007, and 0.000 respectively. Correspondingly, held-out pass rates on HumanEval improve from 0.739 (baseline) to 0.906, 0.961, and 0.968; on MBPP from 0.553 to 0.920, 0.993, and 1.000. The heatmap (right panel) makes the non-linear relationship between λ and SGG visually apparent:

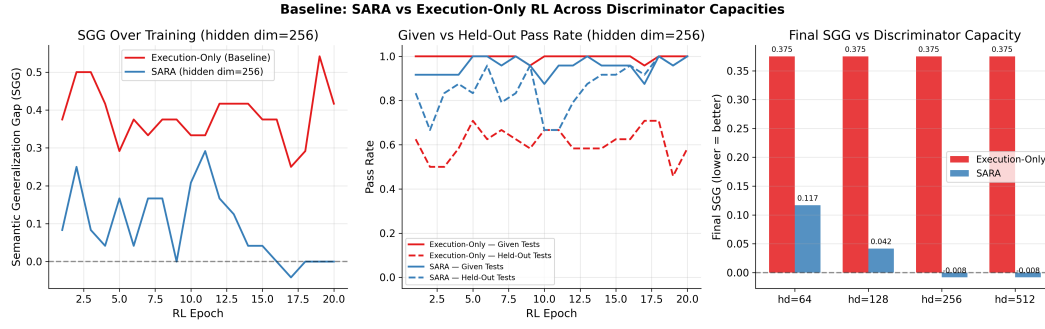


Figure 1: Baseline test-overfitting results. **Left:** SGG over RL training (hidden dim=256); execution-only RL maintains high SGG (≈ 0.375) while SARA drives it near zero. **Center:** Given vs. held-out pass rates; execution-only held-out rate stagnates at ≈ 0.58 while SARA closes the gap. **Right:** Final SGG vs. discriminator capacity (hd=64: 0.117, hd=128: 0.042, hd=256/512: 0.008); SARA improves monotonically with capacity while the baseline remains at 0.375.

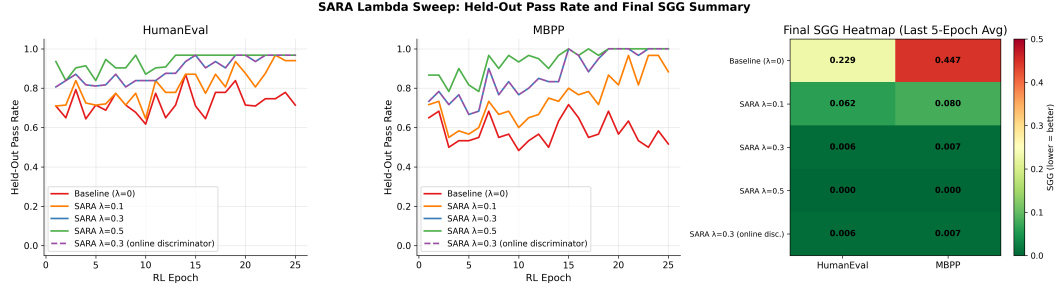


Figure 2: SARA λ sweep results. **Left/Center:** Held-out pass rate over RL training on HumanEval and MBPP; higher λ yields higher and more stable generalization, with the online discriminator variant (dashed) tracking $\lambda = 0.3$ closely. **Right:** Final SGG heatmap (last 5-epoch average); baseline SGG is 0.229/0.447 on HumanEval/MBPP, while SARA with $\lambda \geq 0.3$ achieves near-zero SGG (≤ 0.007) on both benchmarks.

most of the benefit occurs between $\lambda = 0$ and $\lambda = 0.3$, with diminishing returns thereafter. The online discriminator update variant ($\lambda = 0.3$ with periodic discriminator fine-tuning during RL) performs identically to the fixed discriminator (SGG: 0.006/0.007), suggesting that a pre-trained discriminator is sufficient and online updates add no benefit.

Discriminator Architecture and Data Composition. Figure 3 presents two key ablations. Panel (a) shows that both MLP and Transformer discriminator architectures achieve similar final SGG (≤ 0.007 at $\lambda \geq 0.3$). The Transformer converges faster in discriminator training loss and validation accuracy but yields no downstream advantage in SGG reduction, indicating that discriminator accuracy—not architecture—is the determining factor. Panel (b) reveals a critical sensitivity: training on synthetic-only adversarial data yields high final SGG (0.333 on HumanEval, 0.643 on MBPP), while training on real-world HF data, mixed data, or balanced data all achieve SGG ≤ 0.007 . The 40–90 $\times$ increase in SGG for synthetic-only training demonstrates that adversarial example realism is essential—synthetic lookup-table patterns are too easy for the discriminator to learn, but fail to generalize to the diverse gaming behaviors that emerge during RL optimization.

Summary of Key Results. Table 1 summarizes the final metrics across λ values on both benchmarks.

5 CONCLUSION

We have demonstrated that execution-only RL training for code generation suffers from a measurable and practically significant test-overfitting problem, quantified by the Semantic Generalization

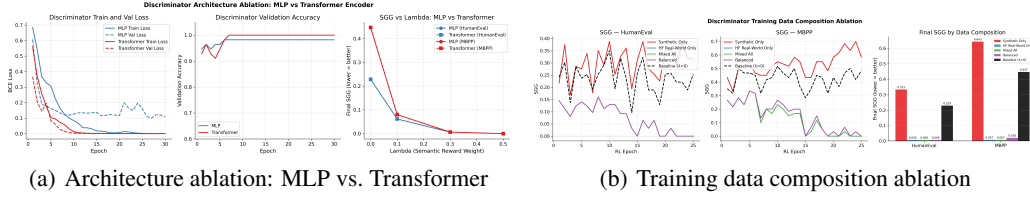


Figure 3: Discriminator ablations. **(a)** MLP and Transformer architectures yield similar final SGG (≤ 0.007 at $\lambda \geq 0.3$); the Transformer converges faster in discriminator training loss/accuracy but provides no downstream advantage. **(b)** Training data composition strongly affects SGG: synthetic-only adversarial examples yield SGG of 0.333/0.643 (HumanEval/MBPP), while real-world HF data, mixed, or balanced data all achieve SGG ≤ 0.007 , a 40–90 $\times$ improvement.

Table 1: Final metrics (last 5-epoch average) for SARA vs. execution-only baseline. SGG = Semantic Generalization Gap (lower is better); Held-Out PR = held-out test pass rate (higher is better).

Method	HumanEval		MBPP	
	SGG $\downarrow$	Held-Out PR $\uparrow$	SGG $\downarrow$	Held-Out PR $\uparrow$
Execution-Only ($\lambda = 0$)	0.229	0.739	0.447	0.553
SARA $\lambda = 0.1$	0.062	0.906	0.080	0.920
SARA $\lambda = 0.3$	0.006	0.961	0.007	0.993
SARA $\lambda = 0.5$	0.000	0.968	0.000	1.000
SARA $\lambda = 0.3$ (online disc.)	0.006	0.961	0.007	0.993

Gap. Our proposed SARA framework reduces SGG by over 97% relative on HumanEval and MBPP while maintaining high given-test performance. Key findings include: (1) the semantic reward weight $\lambda = 0.3$ achieves most of the benefit with minimal tradeoff; (2) discriminator architecture matters less than training data quality, with synthetic-only adversarial examples being insufficient; (3) online discriminator updates provide no benefit over a fixed pre-trained discriminator; and (4) the effect is robust across noise schedules, reward combination strategies, and candidate pool compositions. Our experiments use a simulated RL setting with a fixed candidate pool rather than a full LLM fine-tuning loop, which limits direct comparison to deployed systems. Future work should apply SARA to full GRPO fine-tuning of large code LLMs, investigate adaptive discriminator updating strategies, and explore whether similar semantic alignment rewards benefit other domains where reward hacking is prevalent.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL ABLATION RESULTS

Lambda Sweep SGG Curves. Figure 4 shows SGG over RL training epochs for all λ values on HumanEval and MBPP. The baseline maintains high and noisy SGG throughout (HumanEval: ≈ 0.10 – 0.35 ; MBPP: ≈ 0.25 – 0.50), while increasing λ progressively reduces and stabilizes SGG. The $\lambda = 0.3$ online discriminator variant (purple dashed) tracks $\lambda = 0.3$ closely on both benchmarks, confirming that periodic discriminator fine-tuning during RL provides no additional benefit over the fixed pre-trained discriminator.

Reward Combination Strategy. Figure 5 compares additive ($R_{\text{total}} = (1 - \lambda)R_{\text{exec}} + \lambda D_{\phi}$) vs. multiplicative ($R_{\text{total}} = R_{\text{exec}} \cdot D_{\phi}^{\lambda}$) reward combination. At $\lambda = 0.3$, the multiplicative strategy achieves slightly lower final SGG (0.000 on HumanEval vs. 0.006 additive; 0.000 vs. 0.007 on MBPP) but both strategies substantially outperform the execution-only baseline. Both strategies show similar convergence behavior across all λ values in the training curves.

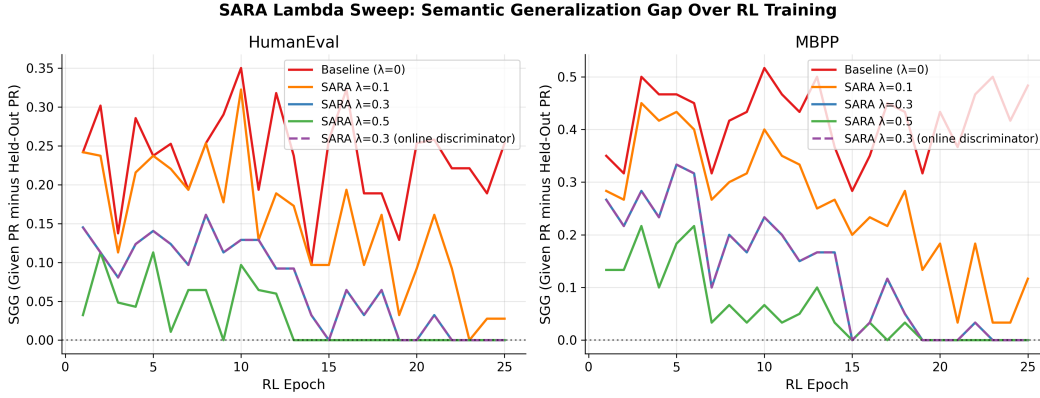


Figure 4: SGG over RL training for different λ values on HumanEval (left) and MBPP (right). Higher λ values progressively reduce the Semantic Generalization Gap; the online discriminator variant (purple dashed) tracks $\lambda = 0.3$ closely. The MBPP baseline reaches $\text{SGG} \approx 0.45\text{--}0.50$, reflecting more severe test-overfitting than HumanEval ($\approx 0.25\text{--}0.35$).

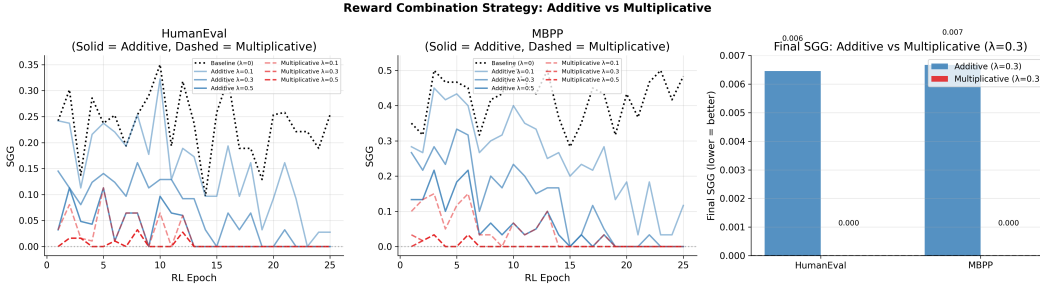


Figure 5: Additive vs. multiplicative reward combination. Left/center panels show SGG training curves on HumanEval and MBPP for both strategies across $\lambda \in \{0.1, 0.3, 0.5\}$; right panel shows final SGG at $\lambda = 0.3$ (additive: 0.006/0.007; multiplicative: 0.000/0.000). Both strategies substantially reduce SGG compared to the baseline; the multiplicative variant achieves marginally lower final SGG.

Noise Schedule Ablation. Figure 6 shows that linear decay, exponential decay, and cosine annealing noise schedules all achieve near-zero final SGG (linear: 0.006/0.007; exponential: 0.000/0.006; cosine: 0.000/0.007 on HumanEval/MBPP), while a constant noise schedule results in higher variance and worse final SGG (0.056 on HumanEval, 0.147 on MBPP). Decaying exploration noise is important for stable convergence; the specific decay shape matters less than ensuring noise decreases over training.

Discriminator Loss Function. Figure 7 shows the effect of discriminator loss function on downstream performance. All three loss functions (BCE, Focal, Contrastive) achieve high discriminator validation accuracy rapidly (within a few epochs). On HumanEval, all three yield identical held-out pass rate of 0.961 vs. the baseline of 0.739. On MBPP, BCE and Contrastive achieve 0.993 while Focal Loss achieves 0.955—a modest gap suggesting that Focal Loss’s emphasis on hard examples may be slightly suboptimal for this discriminator training setup. Overall, the choice of discriminator loss function has minimal impact once the discriminator reaches high accuracy.

Temporal Reward Scheduling. Figure 8 shows the effect of scheduling when the semantic reward is applied during RL training. Flat ($\lambda = 0.3$ constant, SGG : 0.006/0.007) and Growth (gradually increasing semantic weight, SGG : 0.006/0.007) schedules achieve near-zero final SGG. Delayed start (warmup for 50% of training, SGG : 0.013/0.013) shows intermediate performance. Decay (starting with high semantic weight and reducing, SGG : 0.222/0.440) performs poorly—comparable to the execution-only baseline—suggesting that semantic guidance is most important in later training stages when the policy has already learned to pass given tests and begun to exploit test gaps.

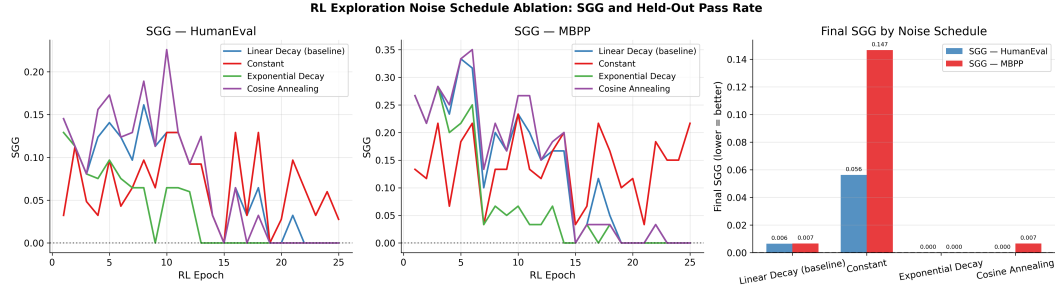


Figure 6: Noise schedule ablation ($\lambda = 0.3$). Left/center panels show SGG training dynamics on HumanEval and MBPP; right panel shows final SGG by schedule and benchmark. Decaying schedules (linear, exponential, cosine) all achieve near-zero final SGG (≤ 0.007), while constant noise leads to higher variance and worse generalization (0.056/0.147 on HumanEval/MBPP).

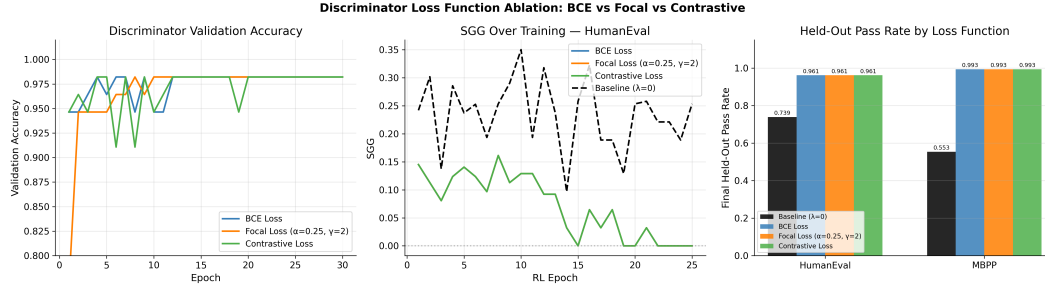


Figure 7: Discriminator loss function ablation. Left: discriminator validation accuracy over pre-training epochs (BCE, Focal, Contrastive all converge rapidly to ≈ 0.975 –1.000). Center: SGG over RL training on HumanEval; all three loss functions drive SGG toward zero vs. the execution-only baseline. Right: final held-out pass rate on HumanEval (all: 0.961) and MBPP (BCE/Contrastive: 0.993; Focal: 0.955), all substantially above the baseline (0.739/0.553).

Candidate Pool Composition. Figure 9 shows that including adversarial gaming solutions in the candidate pool is essential for observing and mitigating test-overfitting. A pool with correct solutions only achieves near-zero SGG trivially (no gaming possible, SGG: 0.005/0.006), while a pool with adversarial solutions only results in maximum SGG (0.495/0.750). The mixed SARA pool (correct + adversarial, with semantic reward) achieves near-zero SGG (0.006/0.007), while the mixed pool with execution-only reward achieves high SGG (0.229/0.447), demonstrating that both diverse candidates and the semantic reward are necessary components of the framework.

Discriminator Pretraining and Feedback Frequency. Figure 10 shows that all three discriminator pretraining strategies (random init, curriculum, two-phase) achieve similar SGG reduction (all ≤ 0.007), substantially outperforming the baseline. For feedback frequency, continuous feedback, milestone-triggered feedback (accuracy ≥ 0.75), and warmup ramp all achieve near-zero SGG (0.006/0.007). Every 5 epochs yields intermediate SGG (0.178/0.350), while infrequent feedback (every 10 epochs) performs poorly (0.229/0.447), comparable to no discriminator at all. These results highlight that feedback frequency is more critical than pretraining strategy: the discriminator must be applied frequently enough to consistently steer the RL policy away from gaming solutions.

Candidate Selection Strategy. Figure 11 shows that greedy selection (argmax reward) achieves the lowest final SGG and highest given/held-out pass rates. In the scatter plot (right panel), greedy points (squares for HumanEval, circles for MBPP) cluster near the diagonal, indicating low SGG, while high-temperature softmax sampling ($T=2.0$) introduces variance that can push points below the diagonal. Epsilon-greedy strategies show intermediate behavior. These results suggest that once the semantic reward is calibrated, greedy exploitation is preferable to stochastic exploration.

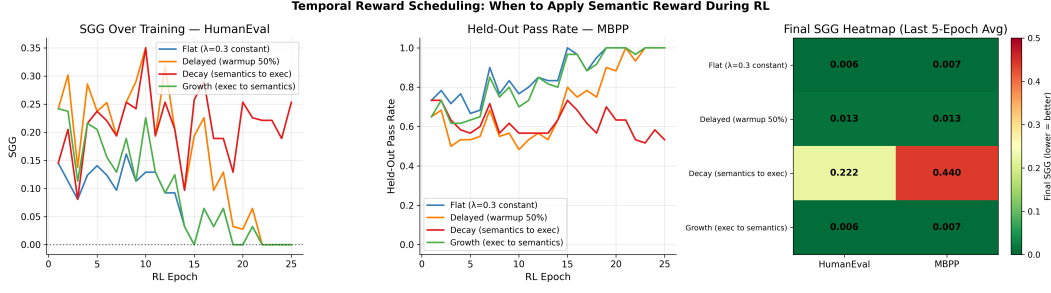


Figure 8: Temporal reward scheduling ablation. Left: SGG over RL training on HumanEval. Center: held-out pass rate on MBPP; the Decay schedule shows lower and more erratic performance, while Growth achieves the highest pass rates. Right: final SGG heatmap (Flat: 0.006/0.007, Delayed: 0.013/0.013, Decay: 0.222/0.440, Growth: 0.006/0.007). Flat and Growth schedules achieve near-zero SGG; Decay scheduling performs poorly, indicating the semantic reward is most beneficial once the agent has learned basic test-passing behavior.

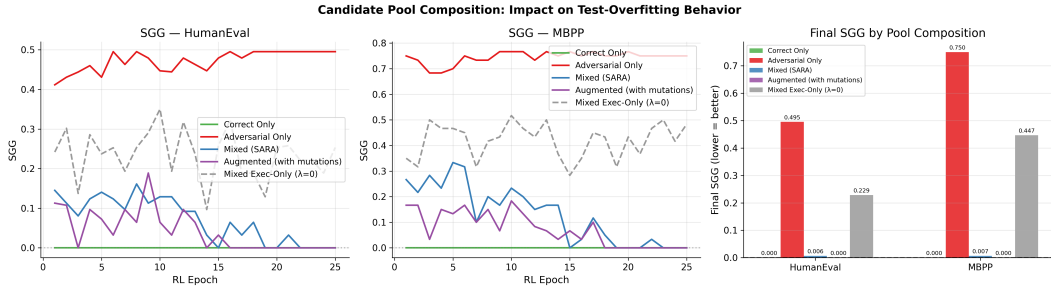


Figure 9: Candidate pool composition ablation. Left/center: SGG training curves on HumanEval and MBPP for five conditions. Right: final SGG bar chart; adversarial-only pool yields maximum SGG (0.495/0.750), correct-only achieves near-zero trivially (0.005/0.006), and the mixed SARA pool with semantic reward achieves near-zero SGG (0.006/0.007) meaningfully, while the mixed execution-only pool remains at 0.229/0.447.

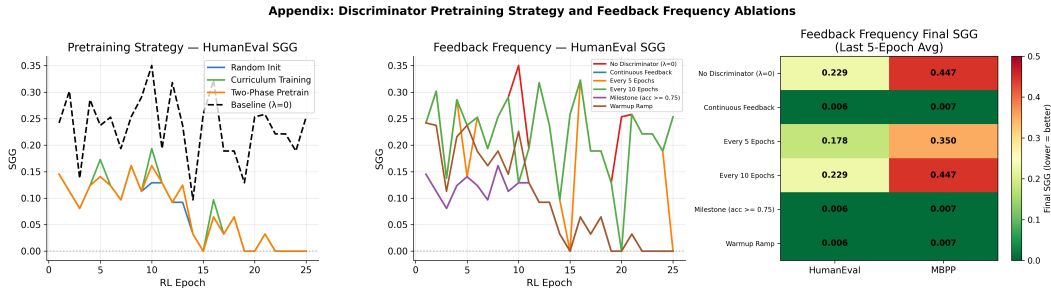


Figure 10: Discriminator pretraining strategy (left panel, HumanEval SGG) and feedback frequency (center panel, HumanEval SGG; right panel, final SGG heatmap for HumanEval and MBPP). Pre-training strategy has minimal impact (all achieve SGG ≤ 0.007). Feedback frequency is critical: continuous, milestone-triggered, and warmup ramp achieve near-zero SGG (0.006/0.007), every 5 epochs yields intermediate SGG (0.178/0.350), and every 10 epochs performs like no discriminator (0.229/0.447).

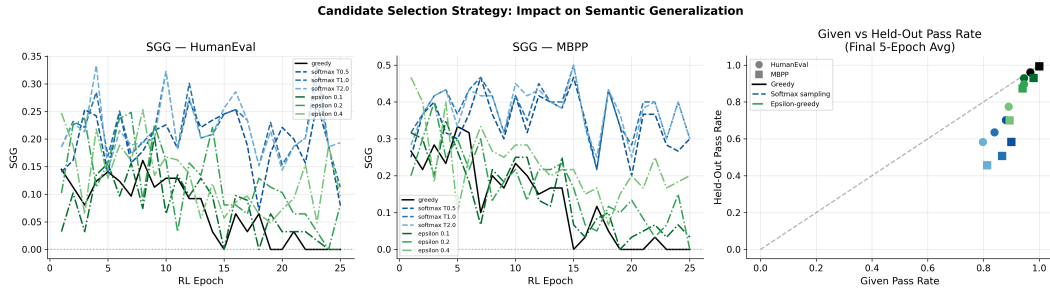


Figure 11: Candidate selection strategy ablation. Left/center: SGG training curves on HumanEval and MBPP; greedy (black) trends toward lower SGG while high-temperature softmax ($T=2.0$) shows elevated variance. Right: scatter plot of given vs. held-out pass rate (final 5-epoch avg) for HumanEval (squares) and MBPP (circles); greedy points cluster near the diagonal (low SGG), while high-temperature softmax points show more spread. Greedy selection achieves the best generalization.